AI DRAWING TAKE-OFF BENCHMARK

One construction drawing and one fixed prompt, run on 17 AI models. This guide sets out the method and the results, and gives you the prompt to run it yourself.

Best average miss per apartment	Window widths read exactly	Within 7% on floor area
0.25%	**483 of 509**	**13 of 17**
Fable 5.2 (High), 1st of 17	Across all 17 models	The worst answer was 1,221.5 m², against 639.1

The test one A1 sheet, the same for every model

1 One sheet	2 Answers out	3 Image only	4 Marked
A.103 from 752 High St at 1:100. Levels 01 and 02, 9 apartments.	The architect's schedules are removed and the printed areas whited out.	The sheet is rasterised, so no text layer gives the answers away.	Each reply is marked against the architect's own schedule.

WHAT EACH MODEL IS ASKED

Task	What counts	How it is marked
Window count	Windows on each level. Doors are left out, including glazed and sliding doors.	Windows found, against 38 in the schedule
Window widths	The width of each window as dimensioned on the plan, in mm.	Widths that match the schedule exactly
Floor areas	Each apartment to the inside face of its walls. Partitions and cupboards in. Decks, balconies and common areas out.	Average miss per apartment

Every run starts fresh, with no project folder or earlier chat, and the prompt fixes the output format: two tables, then the counts and the total.

HOW IT IS SCORED

> The score is the average miss per apartment, not the error on the total. A model 5% over on one apartment and 5% under on another gets the total right and still has a 5% average miss. Misses that scatter over and under mean the apartments need measuring again. A steady miss in one direction is a calibration you can correct.

For each apartment, take the difference from the schedule as a percentage of the schedule figure. Ignore the sign. The score is the mean of the 9.

Where the plan and the schedule disagree, mark against the plan, because the plan is all the model saw.

FABLE 5.2 ON THE TEST

> 0.25% average miss. Its worst apartment was 0.5 m² out, and 2 apartments were exact. It found 29 windows, leaving out the 9 full-height openings as doors, and 27 of the 29 widths were exact. Fable 5.1 scored 2.5% on the same sheet, ten times the miss. It ran on High effort, like every Claude run on the board. Medium is now the default in the Claude app.

THE LEADERBOARD

Ranked best to worst on average miss per apartment.

Rank	Model	Windows found real answer 38	Widths exact of those listed	Floor area real answer 639.1 m²	Average miss per apartment
1	**Fable 5.2 (High)**	29	27 of 29	639.6 m²	0.25%
2	GPT-6 Astra *	28	24 of 24	4 of 9 apartments	0.3% *
3	Fable 5	34	30 of 34	652.6 m²	2.1%
4	Fable 5.1	38	34 of 38	652.5 m²	2.5%
5	GPT-6 Luna	29	25 of 25	645.6 m²	3.0%
6	Sol 6	29	24 of 25	662.7 m²	3.6%
7	Opus 5.5	29	25 of 29	661.4 m²	3.9%
8	Grok Bot app	29	25 of 29	661.6 m²	3.9%
9	Gemini 4 Argon, API with code	29	25 of 29	660.0 m²	4.1%
10	GPT-6.1 Sol	29	25 of 25	650.5 m²	4.1%
11	GPT-5.6 Sol	38	36 of 38	650.5 m²	4.9%
12	Sonnet 5.5	38	34 of 34	674.1 m²	6.4%
13	Opus 5	38	34 of 34	681.7 m²	6.8%
14	Grok 4.7 (Expert), Grok app †	40	26 of 27	700.5 m²	15.2%
15	Haiku 5.5 **	37	32 of 32	633.1 m² (8 of 9)	27.3% **
16	Sonnet 5	40	32 of 32	1,221.5 m²	114%
17	Gemini	30	25 of 25	Would not give one	n/a

* Answered 4 of 9 apartments and refused the 5 where the boundary was unclear. ** Gave 8 of 9 apartments as outside sizes, and said they were not floor areas. † Said 40 windows (18 and 22) but listed 27. API with code: the same sheet and prompt through the Gemini API with code execution on, which the Gemini app does not have.

WHAT IT SHOWS

Printed dimensions are read almost perfectly

Across the 17 models, 483 of 509 window widths matched the schedule. Fable 5.2's 2 misses and Opus 5.5's 4 were all windows with no figured width on the plan, which the model had to scale.

The models split where judgement is needed

The schedule lists 38 windows, and 9 of them are full-height openings onto the decks, most of them sliders. The prompt leaves doors out, so each model decides what those openings are: 7 said 29, 4 said 38, and the other 6 gave counts from 28 to 40. Areas split on where a floor stops. Fable 5.2 noted that the window recesses alone change five apartments by 3.2 to 5.0 m² each.

Stated confidence often does not match the result

Across the runs, stated confidence did not track accuracy, and for some models it ran the wrong way. Fable 5.2 was an exception: the 2 widths it missed were the 2 it said it was least sure of, because the plan gives no figure for them.

The take-off found errors in the drawing

The set was issued FOR CONSTRUCTION, and the models found 5 errors in it. Fable 5.2's reply names examples: a dimension labelled with the wrong window number, and a dimension string beside Apartment 7 labelled Apartment 8.

RUN IT YOURSELF

1. Pick a drawing whose set carries its own schedules. Take the schedules out and white out any answers printed on the plan, so the designer's figures become the marking key. If your set has no schedules, mark against a take-off you have already checked.
2. Turn the sheet into an image if it has a text layer. White boxes hide figures from the eye, but a model can still read the text underneath.
3. Start a fresh chat for each model, with no project folder. Set effort to High.
4. Attach the drawing and paste the prompt below. Replace each marked field with your drawing's details. If you are not measuring apartments, change "apartment" to what you are measuring, such as a unit or an office floor.
5. Score the average miss per item, as on page 1.

The prompt

The test used this prompt with the details of sheet A.103 in the fields.

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You are a quantity surveyor taking off quantities from an
architectural drawing.

These are [issue status, e.g. FOR CONSTRUCTION]
architectural drawings for [building type, e.g. an
apartment building].

**The drawing.** Sheet [sheet number], at [scale] on an
[sheet size] sheet. It contains [number] floor plans: **
[level name]** (apartments [first] to [last]) and **[level
name]** (apartments [first] to [last]). [number]
apartments in total.

Take off the following. Work only from what is shown on
the drawing.

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### 1. Window count

How many **windows** are there on each level?

Count windows only. Do not include doors of any kind,
including glazed doors and sliding doors. If you are
unsure whether an opening is a window or a door, say which
ones you were unsure about and how you decided.

### 2. Window widths

List every window with its ID and its width in
millimetres.

Width means the width of the window itself as dimensioned
on the plan, not the width of the room or the structural
bay.

### 3. Apartment floor areas

Give the floor area of each apartment in m².

**Measure it this way, exactly:**

- Measure to the **inside face of the walls enclosing the
apartment**, the apartment's own side of the wall, not the
middle of the wall and not the outside face.
- **Include** internal partition walls, cupboards, and
every room inside the apartment (this is a gross internal
area, not a sum of room areas).
- **Exclude** decks, balconies, terraces and any external
space.
- **Exclude** common areas outside the apartment: the
foyer, lift, stairs and corridors.

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## Rules

- If you cannot determine a value, say so. Do not estimate
silently.
- State your confidence in each number, and say which
figures you are least sure about.

## Give your answer in exactly this format

**Table 1: Windows.** One row per window. Two columns
only, in this order:

| Window ID | Width (mm) |
|---|---|

Use the window ID exactly as it is tagged on the drawing.
Width as a whole number of millimetres, no units in the
cell, no ranges.

**Table 2: Apartments.** One row per apartment. Two
columns only, in this order:

| Apartment | Floor area (m2) |
|---|---|

Apartment as a number. Area to one decimal place, no units
in the cell.

**Then, below the tables**, state on separate lines:

- Window count for each level
- Total apartment floor area

Put any caveats, assumptions and confidence notes after
those lines, not inside the tables.
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> **Use AI to support a take-off, and do not rely on it alone.** Even the best run on this sheet missed 2 window widths. Check every quantity an AI gives you against the drawing before it goes into a price or a bill.

GET THE DRAWING TAKE-OFF SKILL

Construction AI Accelerator members get a take-off skill for their own drawings. It gives a confidence and a source for every figure, and flags drawing errors.

JOIN THE COMMUNITY